

SURFACES IN E^3 *

by

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We present a brief summary, with references, of developments during the last fifteen years that are related to tame embeddings of 2-manifolds in 3-manifolds. This is the basis of a twenty minute talk presented in the Special Session on Recent Developments in Topology at the meeting of the American Mathematical Society at Iowa City on November 27, 1965. For simplicity in the presentation, we restrict our discussion to 2-spheres in E^3 , but most of the theorems can be extended to 2-manifolds in a 3-manifold.

We have included an extensive list of references on this topic, but we presume that there are some serious omissions and some lack of current information in this list. We find it necessary to place some restriction on the number of papers listed here, and we suggest that other work essential to this development can be found in the references cited in the papers that are included in this list. For example, see Harrold's expository paper [37].

1. Fundamental work related to tame imbeddings of 2-spheres in E^3 .
A 2-sphere S is defined to be tame in E^3 if there is a homeomorphism h of E^3 onto itself that carries S onto the surface of a tetrahedron. Alexander [1] showed in 1924 that every polyhedral 2-sphere in E^3 is tame under this definition. Graeub [30] and Moise [41, II], working independently, showed in 1950 that every polyhedral sphere in E^3 can be carried onto the surface of a tetrahedron with a piecewise linear homeomorphism of E^3 onto itself.

A fundamental characterization of tame spheres in E^3 was given by Bing in 1959 with the following theorem. (In the various theorems on characterizations of tame surfaces in E^3 , we do not include the obvious statement that every tame sphere satisfies the requirements of the characterization.)

1.1 A 2-sphere is tame if it can be homeomorphically approximated in each of its complementary domains. (Bing [10].)

The proof of this theorem depended upon Bing's Approximation Theorem for Spheres [8], which was later generalized to the following Side Approximation Theorem.

* This paper was not presented at the seminar, but it is included by invitation since it treats recent developments and questions discussed at the seminar. This work was supported by NSF Grant GP-3882.

1.2 If S is a 2-sphere in E^3 and $\epsilon > 0$, then there exist (1) a polyhedral 2-sphere S' , (2) a finite collection $D_1, D_2, \dots, D_n$ of disjoint ϵ -discs on S' , and (3) a finite collection of disjoint ϵ -discs $E_1, E_2, \dots, E_n$ on S such that

- (1) S' is homeomorphically with ϵ of S ,
- (2) $S' - \sum D_i \subset \text{Int } S$, and
- (3) $S - \sum E_i \subset \text{Ext } S'$.

Furthermore, $\text{Int } S$ and $\text{Ext } S'$ can be replaced with $\text{Ext } S$ and $\text{Int } S'$, respectively, in conclusions (2) and (3) above. (Bing [14], [17].)

Essential to some of the recent developments on tame surfaces are papers on locally tame sets [7], [41, VIII], triangulation of 3-manifolds [9], [41, V], and Dehn's Lemma and the Sphere Theorem [42], [43]. Brown's work on the generalized Schoenflies Theorem [18] and on locally flat imbeddings [19] is applicable, but for surfaces in E^3 , or in 3-manifolds comparable results can be obtained from the work of Bing and Moise cited above.

2. Examples of wild 2-spheres in E^3 . A 2-sphere in E^3 is defined to be wild if it is not tame. Early examples of wild spheres in E^3 were given by Antoine [4] in 1921 and by Alexander [2] in 1924. The set of wild points in Antoine's example is a wild Cantor set, and in Alexander's example this set is a tame Cantor set. A third type of wild sphere, which has only one wild point, was described by Fox and Artin [27] in 1948. In 1961, Bing [12] described a wild 2-sphere in which every arc is tame. Examples of other types of wild spheres can be found in papers by Alford [3], Ball [5], Bing [6], [16], Cannon [23], Casler [24], Fort [26], Gillman [28], and Martin [40]. Of course other wild spheres can be described by combining some of the methods used in describing these examples.

Relatively little has been done on the equivalence of imbeddings of wild spheres in E^3 . For horned spheres, work by Ball [5] and Casler [24] shows that the horned sphere described by Ball [5] is imbedded differently from the one described by Alexander [2], and Cannon [23] has described a 3-horned sphere that is imbedded differently from either of these. These three examples are free of any knotting in the horns.

3. Other characterizations of tame spheres in E^3 . Let S be a 2-sphere in E^3 . Then S is tame if the requirements of any one of the following statements are fulfilled.

- 3.1 Every disc in S has both the strong enclosure property and the hereditary disc property. (Griffith [31].)
- 3.2 S is locally peripherally unknotted; i.e., for each $p \in S$ and each $\epsilon > 0$, there exists a tame 2-sphere S' such that $p \in \text{Int } S'$, $\text{diam } S' < \epsilon$, and $S \cap S'$ is a simple closed curve. (Harrold, [32], [33].)

- 3.3 The complement of S is 1-ULC. (Bing [11].)
- 3.4 S can be deformed into each complementary domain; i.e., for each component U of $E^3 - S$, there is a homotopy $h: S \times I \rightarrow cl(U)$ such that $h(x, 0) = x$ and $h(x, t) \in U$ for $0 < t \leq 1$. (Hempel [35].)
- 3.5 S is the image of a tame sphere S' under a map f of E^3 onto itself such that $f|S'$ is a homeomorphism and $f(E^3 - S') = E^3 - S$. (Hempel [35].)
- 3.6 S can be locally spanned from each complementary domain; i.e., for each $p \in S$, for each $\varepsilon > 0$, and for each component U of $E^3 - S$, there exist discs D and R such that $p \in \text{Int } R \subset S$, $\text{Int } D \subset U$, $D \cap R = \text{Bd } D = \text{Bd } R$, and $\text{diam } (D + R) < \varepsilon$. (Burgess [21].)
- 3.7 S can be uniformly locally spanned in each complementary domain; i.e., for each component U of $E^3 - S$ and for every $\varepsilon > 0$ there exists a $\delta > 0$ such that for any δ -disc R on S and for any $\alpha > 0$, there is an ε -disc D in U such that $\text{Bd } R$ can be shrunk to a point in $D + (\text{some } \alpha\text{-neighborhood of } \text{Bd } R)$. (Burgess [21].)
- 3.8 (Let K be a Sierpinski Curve; i.e., K is a universal one-dimensional plane curve.) For each component U of $E^3 - S$ and each $p \in S$, there is a homeomorphism $h: K \times I \rightarrow E^3$ such that $h(K \times 0) \subset S$, p is an inaccessible point (relative to S) of $h(K \times 0)$, and $h(K \times t) \subset U$ for $0 < t \leq 1$. (Burgess [21].)
- 3.9 S can be locally spanned in each complementary domain on tame simple closed curves; i.e., for each component U of $E^3 - S$, for each $p \in S$, and for each $\varepsilon > 0$, there exists an ε -disc R in S such that (1) $p \in \text{Int } R \subset S$, (2) $\text{Bd } R$ is tame, and (3) for each $\alpha > 0$ there is an ε -disc D in U such that $\text{Bd } R$ can be shrunk to a point in $D + (\text{an } \alpha\text{-neighborhood of } \text{Bd } R)$. (Loveland [38].)
- 3.10 For each $p \in S$ and each $\varepsilon > 0$ there is a 2-sphere S' satisfying the following requirement: (1) $p \in \text{Int } S'$, (2) $\text{diam } S' < \varepsilon$, (3) $S \cap S'$ is a continuum, and (4) S can be side-approximated missing $S \cap S'$. (Loveland [38].) (See the next section for a definition of the requirement in (4).)

4. Tame subsets of 2-spheres in E^3 . A subset H of a 2-sphere S in E^3 is defined to be tame if H is a subset of a tame sphere. We say that S can be side-approximated missing H if, in the Side Approximation Theorem stated above, it is further required that $H \subset S - \sum E_1$. Bing's methods of using the Side Approximation Theorem to prove the following theorem have been essential in subsequent work on tame subsets of spheres.

- 4.1 If S is a 2-sphere in E^3 and $\epsilon > 0$, then there exists a tame Sierpinski curve K in S such that each component of $S - K$ has a diameter less than ϵ . (Bing [13].)

The following four theorems have been announced in the chronological order in which they are listed, and each of the last three is a generalization of the preceding one.

- 4.2 If the 2-sphere S in E^3 can be side-approximated missing the arc H in S , then H is tame. (Gillman [29].)
- 4.3 If S is a 2-sphere in E^3 , H is a closed subset of S such that the set of all diameters of the components of H has a positive lower bound, and S can be side-approximated missing H , then H is tame. (Hosay [37].)
- 4.4 If the 2-sphere S in E^3 can be side-approximated missing the closed subset H of S and $\epsilon > 0$, then there exists a tame subcontinuum M of S and a null sequence D_i of ϵ -discs in S such that (1) $M = S - \sum \text{Int } D_i$ and (2) $H \subset M - \sum D_i = S - \sum D_i$. (Loveland [39].)
- 4.5 If each of $H_1, H_2, \dots, H_n$ is a closed subset of the 2-sphere S in E^3 and, for each i , S can be side-approximated missing H_i , then $\sum H_i$ is tame and S may be side-approximated missing $\sum H_i$. (Loveland [39].)

Proofs of the following two theorems depend upon the methods used in proving the above four theorems.

- 4.6 If S is a 2-sphere in E^3 , H is a closed subset of S such that the set of all diameters of the components of H has a positive lower bound, and for each $\epsilon > 0$ there is a $\delta > 0$ such that each δ -simple closed curve in $E^3 - H$ can be shrunk to a point in an ϵ -subset of $E^3 - H$, then H is tame. (Hosay [37] and Loveland [39].)
- 4.7 If S is a 2-sphere in E^3 , H is a closed subset of S such that the set of all diameters of the components of H has a positive lower bound, and S can be locally spanned at each point of H from each complementary domain of S , then H is tame. (Loveland [38].)

5. Spheres in E^3 which are tame except on a tame set. Let S be a 2-sphere in E^3 and let H be a tame subset of S such that S is locally tame at each point of $S - H$. We present here some additional requirements on H which imply that S is tame. The foundation for the first result of this type can be found in Moise's work [41, VIII], and this led to the following generalization.

- 5.1 S is tame if H is a tame finite graph. (Doyle and Hocking [25].)

Using consequences of Dehn's Lemma, the Side Approximation Theorem, and his earlier theorem characterizing tame spheres, Bing proved the following more general theorems.

5.2 S is tame if H is a tame Sierpinski curve. (Bing [15].)

5.3 S is tame if H is the union of a finite number of sets each of which is either a tame finite graph or a tame Sierpinski curve. (Bing [15].)

Other results have been obtained where H is the intersection of a tame sphere with S .

5.4 If S is a 2-sphere in E^3 and S' is a tame 2-sphere such that (1) S is locally tame at each point of $S - S'$, (2) $S \cap S'$ does not separate S , and (3) the set of all diameters of components of $S \cap S'$ has a positive lower bound, then S is tame. (Hosay [36].)

5.5 If C is a 3-cell in E^3 and S is a 2-sphere in C such that S is locally tame from $\text{Int } S$ at each point of $S - \text{Bd } C$, then $S + \text{Int } S$ is a 3-cell. (Burgess [20], [22].)

The interior of a disc D in E^3 is defined to be locally tame from one side if for each $p \in \text{Int } D$ there exists a disc D' and a 3-cell E such that $p \in \text{Int } D' \subset D$ and $E \cap D = D' \subset \text{Bd } E$.

5.6 If C is a 3-cell in E^3 and D is a disc in C such that D is locally tame at each point of $D - \text{Bd } C$, then $\text{Int } D$ is locally tame from one side. (Burgess [22].)

5.7 If p is an isolated wild point of a 2-sphere S in E^3 , then S is locally tame at p from one of the components of $E^3 - S$. (Harrold and Moise [34].)

6. Questions on tame spheres and on tame subsets of spheres in E^3 .

The questions raised here have arisen in papers, seminars, and informal discussion in various places. We include a reference where we know a question has appeared in the literature.¹

Which of the following conditions imply that a 2-sphere S in E^3 is tame?

6.1 S can be pierced by a tame disc at each arc in S .

6.2 S can be pierced by a tame annulus at each simple closed curve in S .

6.3 S can be approximated with a map of S in each complementary domain of S [35, p. 280].

6.4 Each Sierpinski curve in S can be pushed with a homotopy (an isotopy) into each component of $E^3 - S$.

¹ Some of these questions were included in an address by R. H. Bing at the International Congress of Mathematicians in 1962. (See pp. 457-548 of the Proceedings of that Congress.)

- 6.5 For each $p \in S$, for each component U of $E^3 - S$, and for each $\epsilon > 0$, there is an ϵ -disc R such that $p \in \text{Int } R \subset S$ and the identity map on $\text{Bd } R$ can be shrunk to a point in an ϵ -subset of $\text{Bd } R + U$ [21, p. 89].
- 6.6 S is locally tame at each point of $S - H$, where H is a non-degenerate tame subcontinuum of S .
- 6.7 S is locally tame at each point of $S - H$, where H is a tame closed subset of S having no degenerate component.
- 6.8 S can be locally spanned in each component of $E^3 - S$.
- 6.9 For each $p \in S$ and each $\epsilon > 0$, there exists a tame 2-sphere S' such that $p \in \text{Int } S'$, $\text{diam } S' < \epsilon$, and $S \cap S'$ is a continuum.
- 6.10 For each $p, q \in S$, there is a homeomorphism of E^3 onto itself that carries S onto itself and p onto q [28, p. 253].
- 6.11 Each horizontal plane that intersects S either intersects S in a point or a simple closed curve [1], [16].

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